3rd Cryptography Homework

4.9.3 Show that the decryption procedures given for the CBC and CFB modes actually perform the desired decryptions.

Decryption procedures:
CBC: $P_j = D_k(C_j) \oplus C_{j-1}$
CFB: $P_j = C_j \oplus L_8(E_k(X_j))$, $X_{j+1} = R_{56}(x_j)||C_j$

4.9.7 Suppose E^1 and E^2 are two encryption methods. Let K_1 and K_2 be keys and consider the double encryption

$$E_{K_1,K_2}(m) = E^1_{K_1}(E^2_{K_2}(m)).$$

4.9.7.a Suppose you know a plaintext-ciphertext pair. Show how to perform a meet-in-the-middle attack on this double encryption.

First, we calculate $E^2_{K_2}(m)$ for all possible K_2. Then, we computes $D^1_{K_1}(E_{K_1,K_2}(m))$ for all possible K_1. Finally, we compares the two lists, there is at least one match.

4.9.7.b An affine encryption given by $x \mapsto \alpha x + \beta \pmod{26}$ can be regarded as a double encryption, where one encryption is multiplying the plaintext by α and the other is a shift by β. Assume that you have a plaintext and ciphertext that are long enough that α and β are unique. Show that the meet-in-the-middle attack from part (a) takes at most 38 steps (not including the comparisons between the lists). Note that this is much faster than a brute force search through all 312 keys.

The first step to calculate $E^2_{K_2}(m)$ for all possible $K_2 = \beta$ takes at most 26 steps and the second step to compute $D^1_{K_1}(E_{K_1,K_2}(m))$ for all possible $K_1 = \alpha$ takes at most $\phi(26) = 12$ steps. Therefore, the mitm attack at most 38 steps.

4.9.8 Suppose we modify the Feistel setup as follows. Divide the plaintext into three equal block: L_0, M_0, R_0. Let the key for the ith round be K_i and let f be some function that produces the appropriate size output. The ith round of encryption is given by

$$L_i = R_{i-1}, M_i = L_{i-1}, R_i = f(K_i, R_{i-1}) \oplus M_{i-1}.$$

This continues for n rounds. Consider the decryption algorithm that start with the ciphertext

$$A_{i-1} = B_i, B_{i-1} = f(K_i, A_i) \oplus C_i, C_{i-1} = A_i.$$

This continues for n rounds, down to A_0, B_0, C_0. Show that $A_i = L_i, B_i = M_i, C_i = R_i$ for all i, so that the decryption algorithm returns the plaintext.

We know that $A_n = Ln, B_n = M_n$, and $C_n = R_n$. $A_{i-1} = B_i = M_i = L_{i-1}$, i.e., $A_i = L_i$ for all i.
$C_{i-1} = A_i = L_i = R_{i-1}$. Hence $C_i = R_i$ for all i.
$B_{i-1} = f(K_i, A_i) \oplus C_i = f(K_i, L_i) \oplus R_i = f(K_i, R_{i-1}) \oplus R_i$, we obtain that

$$R_i = f(K_i, R_{i-1}) \oplus B_i.$$

Hence, $B_i = M_i$ for all i.

4.9.9 Consider the following simplified version of the CFB mode. The plaintext is broken into 32-bit pieces: $P = [P_1, P_2, \cdots]$, where each P_j has 32 bits, rather than the 8 bits used in CFB. Encryption proceeds as follows. An initial 64-bit X_1 is chosen. Then for $j = 1, 2, 3, \cdots$, the following is performed:

$$C_j = P_j \oplus L_{32}(E_K(X_j))$$

$$X_{j+1} = R_{32}(X_j)||C_j,$$

where $L_32(X)$ denotes the 32 leftmost bits of X, $R_{32}(X)$ denotes the rightmost 32 bits of X, and $X||Y$ denotes the string obtain by writing X followed by Y.

4.9.9.a Find the decryption algorithm.

Decryption algorithm:

$$P_j = C_j \oplus L_{32(E_K(X_j))}$$

$$X_{j+1} = R_{32(X_j)||C_j}$$

4.9.9.b The ciphertext consists of 32-bit blocks $C_1, C_2, C_3, C_4, \cdots$ Suppose that a transmission error causes c_1 to be received as $\tilde{C_1} \neq C_1$, but that $C_2, C_3, C_4, \cdots$ are received correctly. This corrupted ciphertext is then decrypted to yield plaintext blocks $\tilde{P}_1, \tilde{P}_2, \cdots$ Show that $\tilde{P}_1 \neq P_1$, but $\tilde{P}_i = P_i$ for all $i \geq 4$. Therefore, the error affects only three blocks of the decryption.

$iv = 64bits$, when $\tilde{C_1} \neq C_1$, $\tilde{P}_1 \neq P_1$. Since $\tilde{X}_2 = R_{32}(X_1)||\tilde{C}_1$ and $\tilde{C}_1 \neq C_1$, we obtain $\tilde{X}_2 \neq X_2$ and $\tilde{P}_2 \neq P_2$. Then, $\tilde{X}_3 = R_{32}(X_2)||\tilde{C}_2$ and $\tilde{C}_1 \neq C_1$, i.e., $\tilde{X}_2 \neq X_2$ and $\tilde{P}_2 \neq P_2$. Last, $\tilde{X}_4 = R_{32}(X_3)||\tilde{C}_3$, $\tilde{C}_1$ will be ignored and we see that $\tilde{X}_4 = X_4$ and $\tilde{P}_4 = P_4$.